

Car Dodging Game

Repo Clone : <https://github.com/codertheashish/Car-Dodging-Game>

1. Introduction

The **Car Dodging Game** is a simple 2D arcade game developed using **Python** and the **Pygame** library. In this game, the player controls a car and avoids enemy cars moving from the top of the screen. The score increases whenever an enemy car passes the player successfully. As the score increases, the enemy cars move faster, making the game more challenging. If the player's car collides with an enemy car, the game ends and a **Game Over** screen is displayed with options to restart or exit.

2. Purpose

The main purpose of this project is to learn the basics of **2D game development** and improve programming skills using Pygame.

This project helps in learning:

- Player movement
 - Object movement
 - Collision detection
 - Score management
 - Random obstacle generation
 - Dynamic difficulty
 - Game restart system
 - Event handling
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3. Technologies Used

Technology	Purpose
Python	Main programming language
Pygame	Creates the game window, graphics, animations, and keyboard controls
Random Module	Generates random positions for enemy cars
Sys Module	Safely closes the game when the player exits

4. Main Features

✓ Player Car Movement

The player controls the yellow car using the keyboard.

Controls:

- **Left Arrow** → Move Left
- **Right Arrow** → Move Right

The car can only move within the game window and cannot cross the screen boundaries.

✓ Enemy Car System

The game starts with **three enemy cars**.

- Enemy cars move from the top to the bottom of the screen.
 - When an enemy reaches the bottom, it is removed and a new enemy appears at a random position at the top.
 - This creates a continuous flow of obstacles throughout the game.
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✓ Score System

The score increases by **1** whenever an enemy car successfully passes the player and leaves the screen.

The current score is displayed in the top-left corner throughout the game.

✓ Dynamic Difficulty

The game becomes more challenging as the player scores more points.

- Every **5 points**, the speed of the enemy cars increases.
 - This makes the game progressively harder and more exciting.
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✓ Collision Detection

The game continuously checks for collisions between the player's car and the enemy cars.

If a collision occurs:

- The game immediately stops.
 - A **Game Over** screen is displayed.
 - The final score is shown to the player.
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✓ Restart and Exit Options

After the Game Over screen appears:

- **Press R** → Restart the game.
- **Press ESC** → Exit the game.

When the game restarts:

- The player car returns to the starting position.
 - New enemy cars are generated.
 - The score resets to **0**.
 - Enemy speed returns to its initial value.
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✓ Smooth Gameplay

The game runs at **60 FPS**, providing smooth movement for both the player and enemy cars.

This creates a better gaming experience with responsive controls and animations.

Summary

"This project is a Car Dodging game developed using Python and the Pygame library. The player controls a car and avoids randomly generated enemy cars. The game includes player movement, obstacle spawning, collision detection, score tracking, automatic speed increase, and a Game Over screen with restart and exit options. Through this project, I learned game loops, object movement, collision detection, random obstacle generation, and event handling using Pygame."